A Project Report on

"Exploratory Data Analysis (D3): Check and Independent Validation"

Group M

Analytics Capstone Project (ANCP) · Westside Stylist Mirror · Module M3

Submitted to

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Repository: https://github.com/cprashanth666666-sys/Captain/tree/prash/D3



EDA validation report: was the analysis done, and does it hold up?

ANCP D3 (lead owner C Prashanth) · checked 3 Nov 2026 · Linked outcomes CO2, CO4 · Repository: https://github.com/cprashanth666666-sys/Captain/tree/prash/D3

1 Verdict

Yes, the EDA was performed. The repository holds a notebook of 57 cells (30 code cells, every one with saved output, 0 errors), 12 figures, an insight log of 10 findings, 7 hypothesis verdicts and a two-page summary. The hypotheses and their pass rules were written before any data was loaded.

Yes, the numbers validate. A clean re-run of the notebook reproduces all 122 key numbers exactly. An independent recomputation from the D2 cleaned data matches 25 of 25 headline figures within tolerance (most to 4 decimal places). All 7 of 7 hypothesis verdicts follow from the stated rules. We rate the EDA **validated for use in M4, with the caveats in section 4**.

2 What is in the repository

Item	Path (relative to the repository)	Purpose
EDA notebook	D3/ANCP_TM_D3_eda.ipynb (and .html)	Questions, hypotheses, cleaning, 12 figures, verdicts, insight log
Insight log	D3/outputs/insight_log.csv	10 findings with evidence, so-what and confidence
Key numbers	D3/outputs/key_numbers.json	122 values the summary quotes
Summary	D3/ANCP_TM_D3_summary.pdf	Two-page decision summary
Reference profile	D3/reference/full_population_profile.json	Full 31.8M-row file profile for the sample check
Validation code and results	D3/validation/validate_eda.py, validation_results.json	This report's checks

The raw and cleaned data files are not listed here: they derive from the Kaggle H&M competition data, which may not be redistributed (see the D2 report).

3 How it was validated

Test	What was done	Result
T1 Runs clean	Read the saved notebook: error outputs and code cells without output.	0 errors; 30 of 30 code cells have output.
T2 Reproduces	Re-executed the whole notebook in a fresh copy against the raw files and compared key_numbers.json value by value.	All 122 values identical (run time excluded).
T3 Independent recount	Rebuilt 25 headline figures from the D2 cleaned tables with new code: visits from customer-days, a different Gini formula, weekly top-100 overlap, colour shares, channel and age splits.	25 of 25 within tolerance (table below).
T4 Verdict logic	Applied the notebook's pre-set thresholds to its own reported numbers for H1 to H7.	7 of 7 verdicts agree.
T5 Text vs numbers	Checked the insight log and the summary PDF quote the same values as key_numbers.json.	All checked values match.

T3 also tests the link between deliverables: D3 cleans the raw files inside the notebook, D2 cleans them with a script. Both give the same 3,066,344 units and 132,435 customers.

3.1 Independent recount against D3 (selected rows; all 25 in validation_results.json)

Figure	D3 notebook	Independent	Gap	Within tolerance
units in scope	3,066,344	3,066,344	0.0000	Yes
customers in scope	132,435	132,435	0.0000	Yes
mean units per visit	3.4551	3.4551	0.0000	Yes
single-unit visit share	0.3011	0.3011	0.0000	Yes
customers with one visit	0.3202	0.3202	0.0000	Yes
Gini of units per customer	0.6425	0.6425	0.0000	Yes
top 10% customers' share of units	0.4804	0.4804	0.0000	Yes
median age (known ages)	31.0000	31.0000	0.0000	Yes
menswear share of units	0.0584	0.0584	0.0000	Yes
top-3 colour share of known-colour units	0.6158	0.6154	0.0004	Yes
weekly units Mar-Aug 2020 vs 2019	-0.1307	-0.1307	0.0000	Yes
channel-2 share Mar-Aug 2020	0.7514	0.7514	0.0000	Yes
single-unit share, age 65+	0.4347	0.4347	0.0000	Yes
garment visits that are complete outfits	0.4502	0.4502	0.0000	Yes
top-100 overlap (Jaccard), 1-week gap	0.2887	0.2898	0.0011	Yes
next-week coverage of last week's top 12	0.0163	0.0163	0.0000	Yes
top 1% of articles' share of units	0.1445	0.1445	0.0000	Yes

4 Caveats and gaps, most important first

1. **H5 verdict and insight I7 pull in different directions.** H5 is "Not supported" because Cramér's V is 0.057 (rule: at least 0.15), yet the single-unit share differs by 17 points between age bands, which passes its own 5-point rule. Both statements are correct under the AND rule, but a reader of the summary could take H5 to mean "no age difference". The summary should say so.

2. **Not independently recomputed.** The exact-article repeat ratio (19×) behind H2, the co-purchase lift behind H3, the confidence intervals, Cramér's V and the target-group proxy were checked only by the clean re-run (T2), not by new code. They are reproducible, not yet cross-checked.

3. **Thresholds are team judgements.** A verdict flips if a threshold moves; for example H1 needs 50% single-unit visits and the data show 30%, so that verdict is robust, but H2 (1.26× against a 1.5× rule) is closer. The notebook does not include a sensitivity table.

4. **Proxy limits remain.** The data is a European retailer's; the sales channel is an unlabelled code; a "visit" is a customer-day because there is no bill ID. I2 (thin baskets in one channel) is rightly rated medium confidence.

5. **EDA, not model evidence.** The notebook uses the whole two-year window. That suits description, but none of its hit rates or overlaps is a model result. M4 must rebuild them behind the date split in the D2 data.

6. **Same sample.** All checks reuse the one 10% sample, so they test the arithmetic and logic, not whether another sample would give the same patterns.

5 Recommended next steps

(1) Add one sentence to the D3 summary explaining H5. (2) Add a threshold-sensitivity table for H2 and H5. (3) In M4, load the D2 cleaned tables (they reproduce D3's numbers) instead of cleaning inside each notebook. (4) Confirm the meaning of sales channel codes 1 and 2 with the course contact before any in-store claim.

6 Reproduce

`python -I D3/validation/validate_eda.py D2/data-v1.0 D3/outputs D3/validation/validation_results.json` recreates T3 and T4. T2 needs the raw Kaggle files and the notebook runner in D3. The checks were written and run with the help of Claude Code; every figure above comes from the saved results file.